

Computational Linguistics 1 (2026)

Assignment 3: Phase 2

(HMM and CRF Sequence Modeling)

Deadline: April XX, 2026 (11:59 PM)

Objective

Using the datasets from Phase 1 and provided gold datasets, you will implement and compare two classic sequence models: **Hidden Markov Models (HMM)** and **Conditional Random Fields (CRF)**.

Task 3: Hidden Markov Model (HMM) Tagger (20 Marks)

Implement an HMM-based sequence tagger.

The model should estimate:

$$P(T|W) = \prod_i P(w_i|t_i)P(t_i|t_{i-1})$$

Where:

- $P(t_i|t_{i-1})$ = transition probability
- $P(w_i|t_i)$ = emission probability

You must implement:

- probability estimation
- smoothing for unknown words
- decoding using the **Viterbi algorithm**

The HMM should be used for:

- POS tagging
- Chunk tagging

Resources provided:

- **Baseline Code:** HMM Google Colab Baseline.
- **Gold Datasets:**
 - UD English-EWT (for English).
 - LTRC Shallow Parsing Data (for Indian Languages).

Note: The provided code is a baseline. This code will require you to **make a few changes** to adapt it for the specific data formats and the requirements of the next part of the assignment.

Task 4: Conditional Random Fields (CRF) Tagger (20 Marks)

Train a CRF model (`sklearn-crfsuite`) for shallow parsing.

- **Data:** Use the gold datasets provided in Task 3 for training and compare results against the LLM-generated data from Task 2.
- Use contextual features such as lowercase word, prefixes/suffixes, and neighboring words.

Use contextual features such as:

- word
- lowercase word
- prefix
- suffix
- is capitalized
- previous word
- next word

You may design additional linguistic features.

Task 5: Evaluation & Error Analysis (20 Marks)

Evaluate both models against the **Phase 1 manual dataset**.

- **Metrics:** POS Accuracy, Chunking F1, NER F1.
- **Error Analysis:** Discuss 5 POS, 5 Chunking, and 5 NER errors. Compare the behavior of HMM vs. CRF vs. LLM annotations.

Submission Instructions

Submit a ZIP file containing:

- `hmm_tagger.py` and `crf_tagger.py`
- `evaluation.py`
- `final_report.pdf` (Results and Error Analysis)